

Ahmed Abdelrehim

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EDUCATION

Hamilton College

Clinton, NY

Bachelor of Arts in Computer Science, Mathematics; GPA: 3.88

Dec. 2024

- Magna Cum Laude; CS Honors; ICPC NENA Regional 1st Undergraduate Team (March 2022)
- Study Abroad: The University of Manchester (Jan. 2023 – Jan. 2024)
- Affiliations: Phi Beta Kappa, ColorStack, CodePath
- Relevant Coursework: Advanced DS&A, Distributed Systems, AI & ML, Deep Learning, Cognitive Robotics

EXPERIENCE

Meta

Sunnyvale, CA

Software Engineer — ML Infrastructure

Feb. 2025 – Present

- Migrated the backend orchestrator for a high-usage ML model iteration workflow for Ads ranking, reducing processing latency by **~88%** and accelerating experiment turnaround for model development
- Developed a generalized machine learning workflow to scale a key ML technique across production Ads models, driving an average **2.45%** accuracy gain across related prediction tasks where deployed
- Designed and implemented a caching layer for a new set of model performance metrics, with projected savings of **~102** engineering hours annually by reducing repeated computation and data retrieval overhead
- Built an end-to-end model analysis and evaluation tool for ad-hoc and automated statistical analysis of Ads models, generating actionable insights to identify performance regressions and model improvement opportunities
- Creating a platform for dynamic workflows to perform architecture-level changes across multiple Ads ML models, handling software and hardware heterogeneity and generate baseline models to speed up iterations by **~2 Wks**
- Working on a vision for an agentic orchestration solution that groups compatible ML techniques within an iteration, using technique metadata, conflict signals, and training resource constraints to maximize model gains

HPE - Juniper Networking

Boston, MA

Software Engineer Intern

May 2024 – Aug. 2024

- Rewrote a Python microservice, removing obsolete functionality and reducing code size by **72%**
- Led migration of Address Pool Manager (APM) monitoring to use Prometheus clients instead of terminal logs
- Created custom metrics endpoints as parallel processes with instrumentation code and gRPC
- Configured Helm chart to scrape metrics from the K8s cluster running APM and endpoints on known ports
- Built Grafana dashboards to view metrics and Redis-stored values for real-time monitoring

Hamilton College R&ID Team

Clinton, NY

Software Developer

June 2023 – Jan. 2024

- Integrated AWS DynamoDB, S3 buckets, and **4** Lambda functions for CRUD operations in an inventory app
- Restructured REST API endpoints in Django, enhancing response times by **14%**
- Revamped UI of a **7-page** digital archive using React after pitching a Figma wireframe
- Automated data collection via Python scripts, populating MySQL DB with **1,500+** objects

Hamilton College Computer Science Department

Clinton, NY

ML Research Assistant

May 2022 – Aug. 2022

- Fine-tuned a Genetic Programming ML model in Clojure, reducing required training cases by **~11%**
- Fixed bugs in evaluation functions for **~20** benchmark tests, increasing success rate by **~10%**
- Automated experiment execution on HPC cluster and results visualization in RStudio with Python

TECHNICAL SKILLS

Programming: Python, C/C++, Golang, PHP, SQL, HTML/CSS, JavaScript, TypeScript, Clojure

ML / Data: PyTorch, scikit-learn, NumPy, Matplotlib, model evaluation, statistical analysis

Backend / Infrastructure: Distributed systems, ML workflow orchestration, GraphQL, REST APIs, gRPC, Docker, Kubernetes (K8s), Redis, MongoDB, AWS

Observability / Tooling: Git, Prometheus, Grafana, PyTest, Linux/Unix, ArgoCD